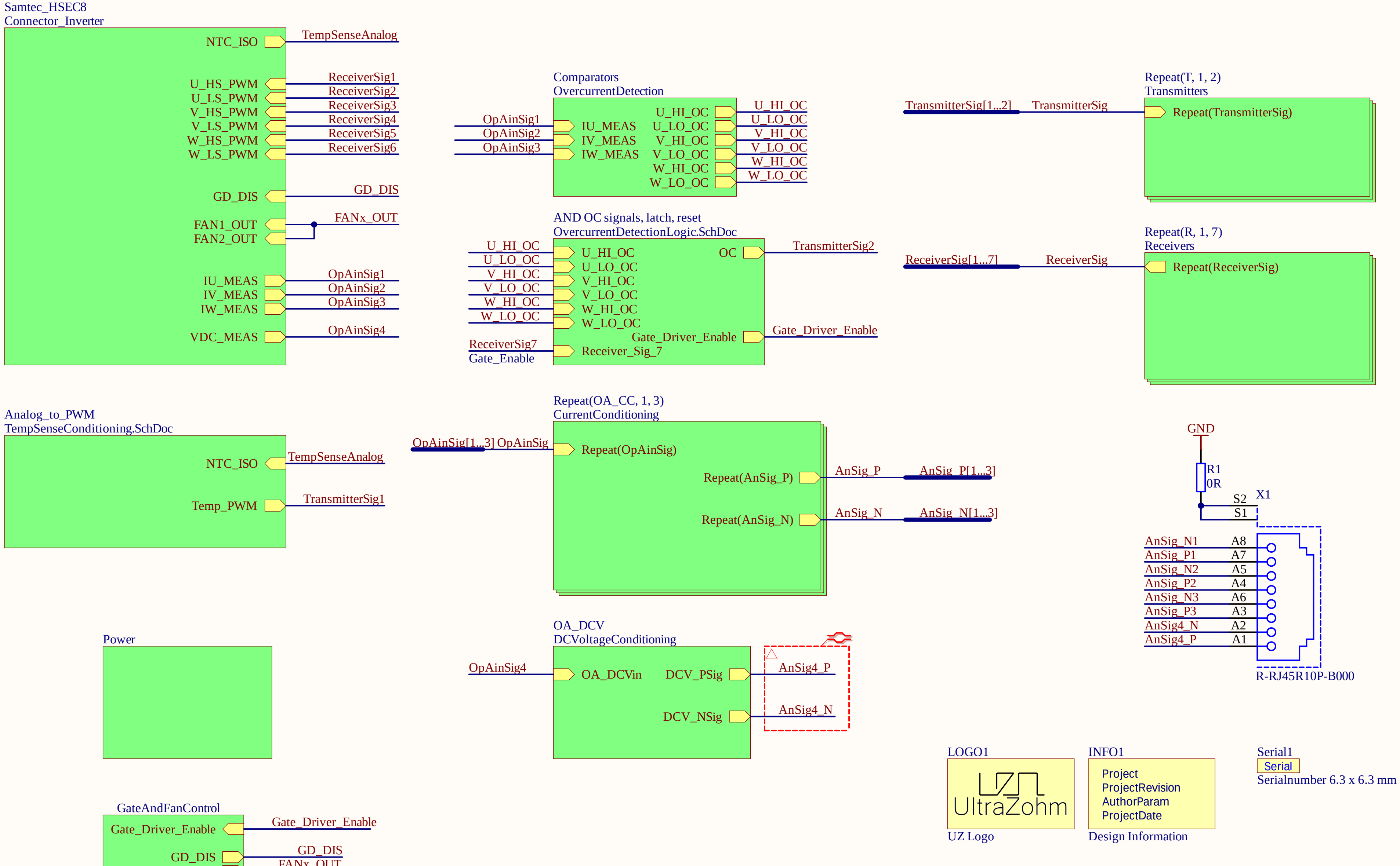
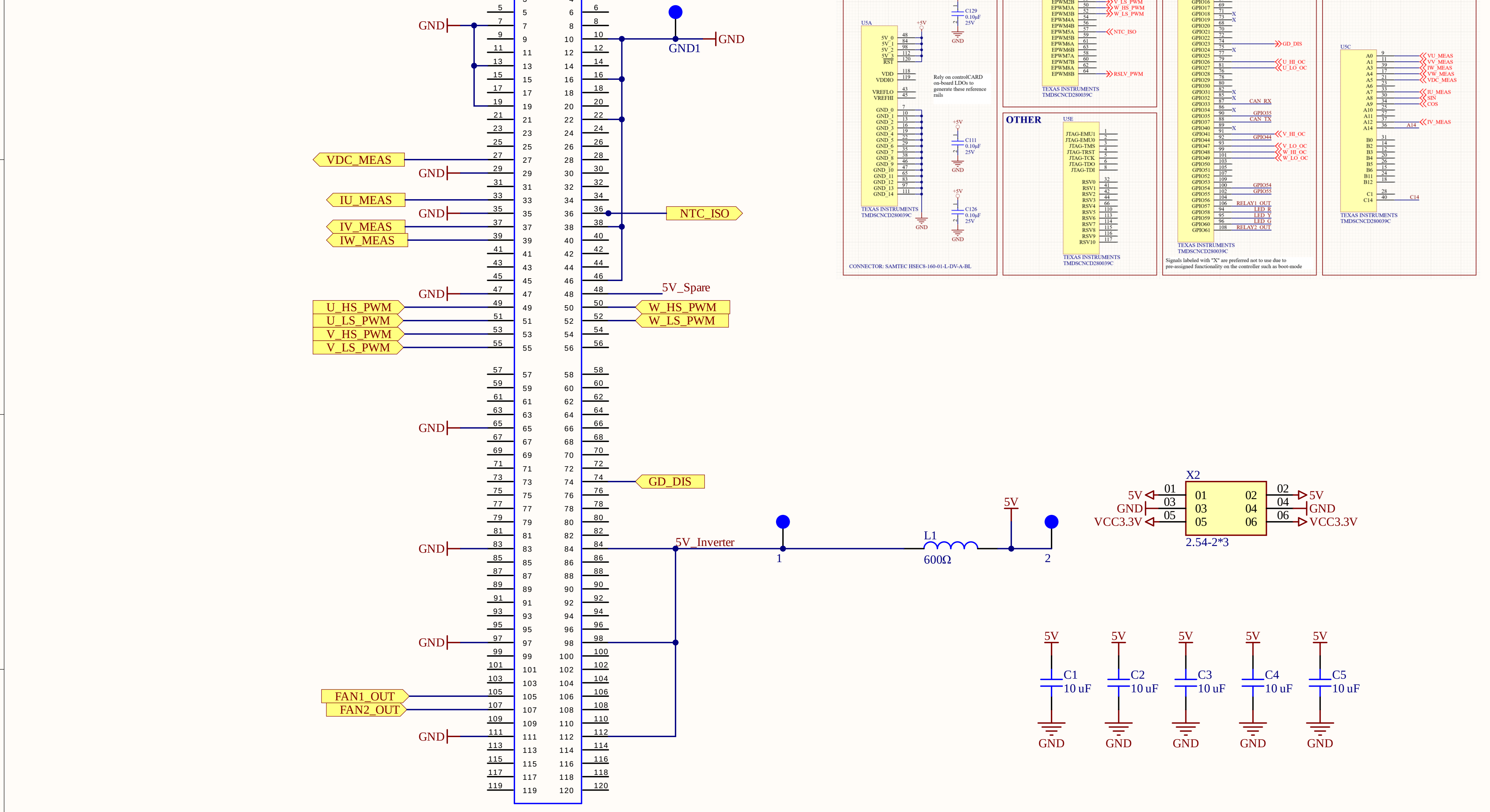
Interface PCB for Wolfspeed CRD25DA12N-FMC 25 kW inverter Rev2.0 ONLY !

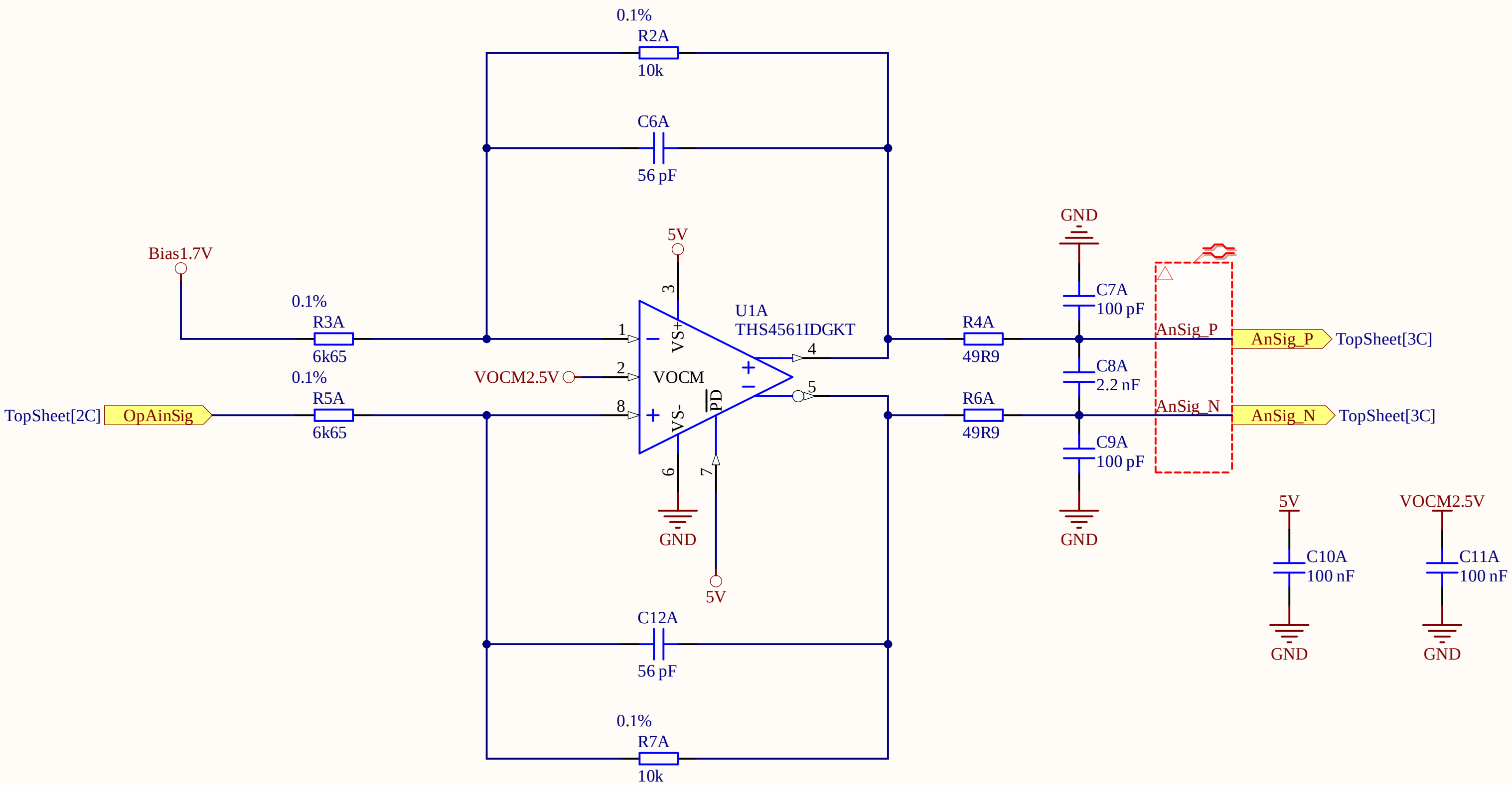


Title TopSheet.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	Date: 23.06.2026
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Sheet 1 of 20

TI Definition: Pin 1 = Pin 120



Title Connector_Inverter.SchDoc	UltraZohm www.ultrazohm.com
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	Date: 23.06.2026
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Sheet 2 of 20



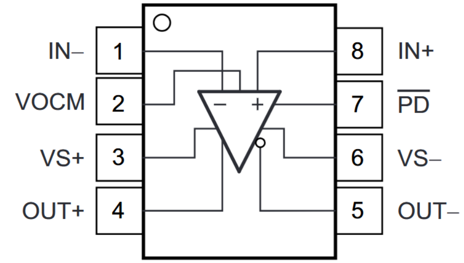
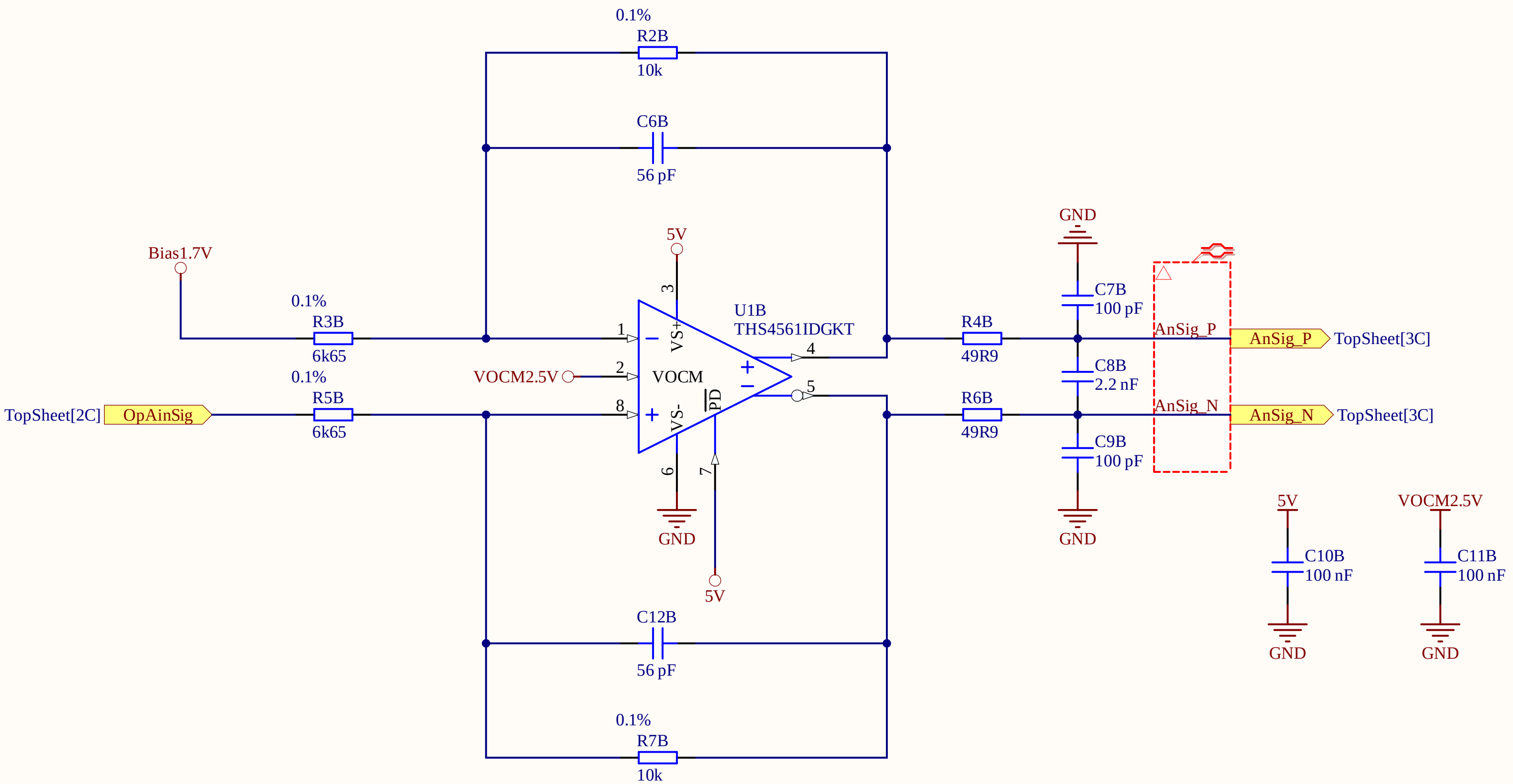


Figure 6-1. DGK Package 8-Pin VSSOP Top View

Title CurrentConditioning.SchDoc		*UltraZohm* www.ultrazohm.com
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026 Sheet 3.1 of 20



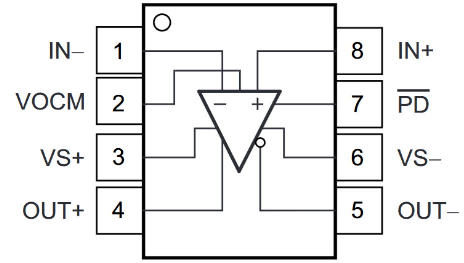


Figure 6-1. DGK Package 8-Pin VSSOP Top View

Title CurrentConditioning.SchDoc	*UltraZohm* www.ultrazohm.com
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Date: 23.06.2026 Sheet 3.2 of 20

1 2 3 4

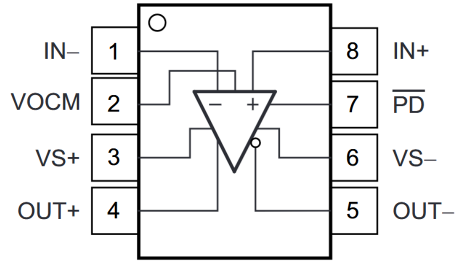
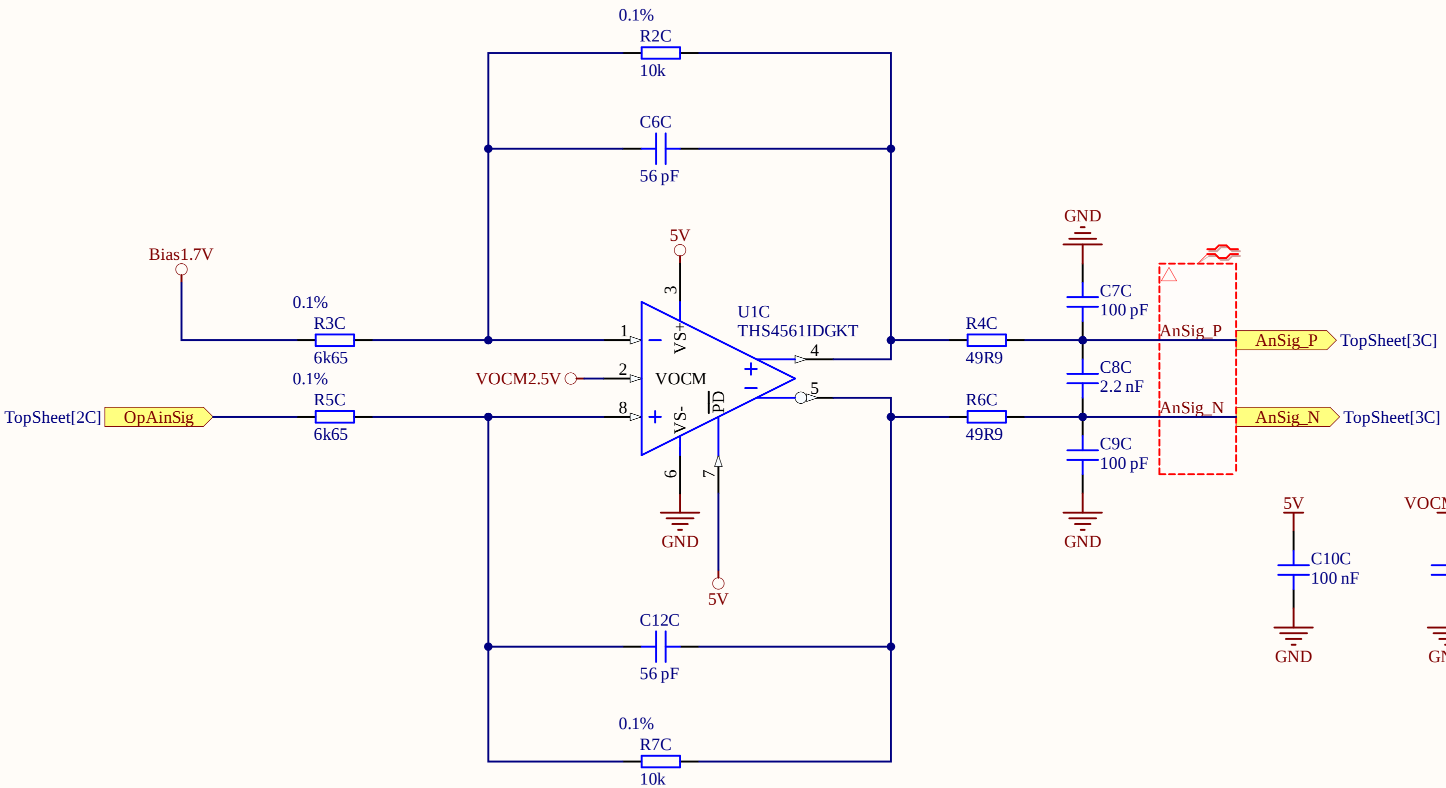


Figure 6-1. DGK Package 8-Pin VSSOP Top View



Title CurrentConditioning.SchDoc		*UltraZohm*
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026 Sheet 3.3 of 20

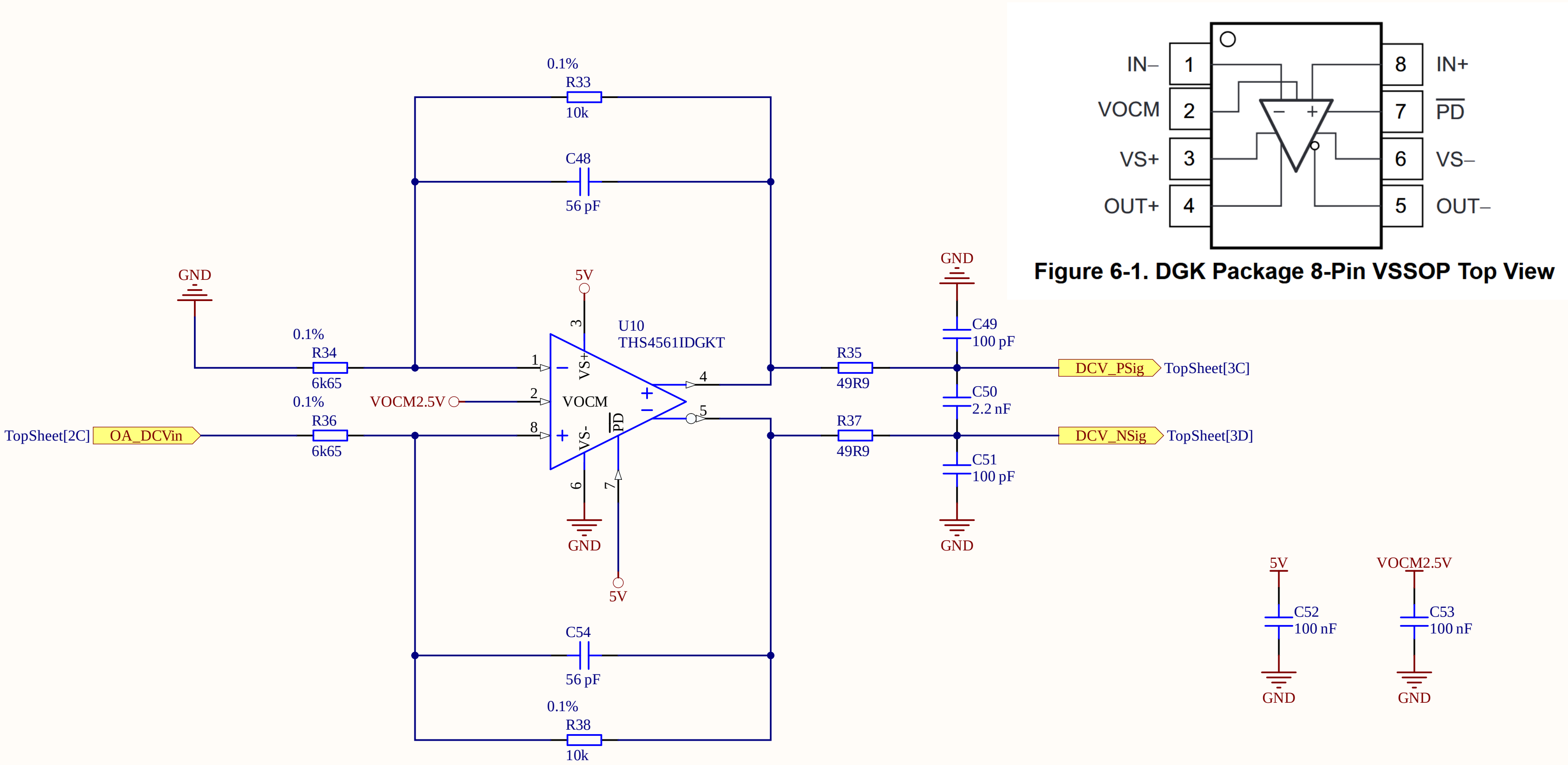
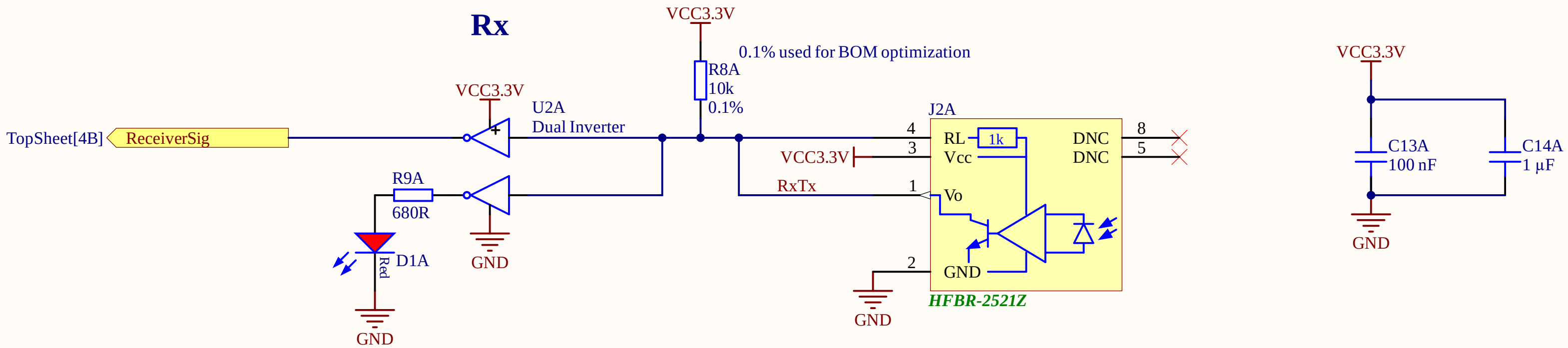


Figure 6-1. DGK Package 8-Pin VSSOP Top View

Title DCVoltageConditioning.SchDoc	UltraZohm www.ultrazohm.com
Revision: Rev04 · Design Engineer: S. Park/M. Hoerner	Date: 23.06.2026
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Sheet 4 of 20

Poor Mans Table of Truth for Rx

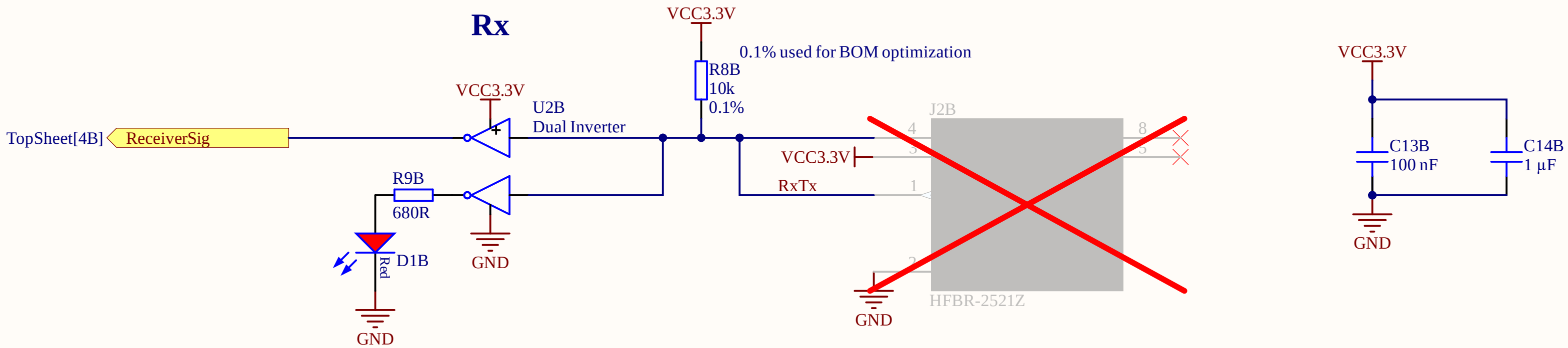
Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Receivers.SchDoc	UltraZohm
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Date: 23.06.2026 Sheet 5.1 of 20

Poor Mans Table of Truth for Rx

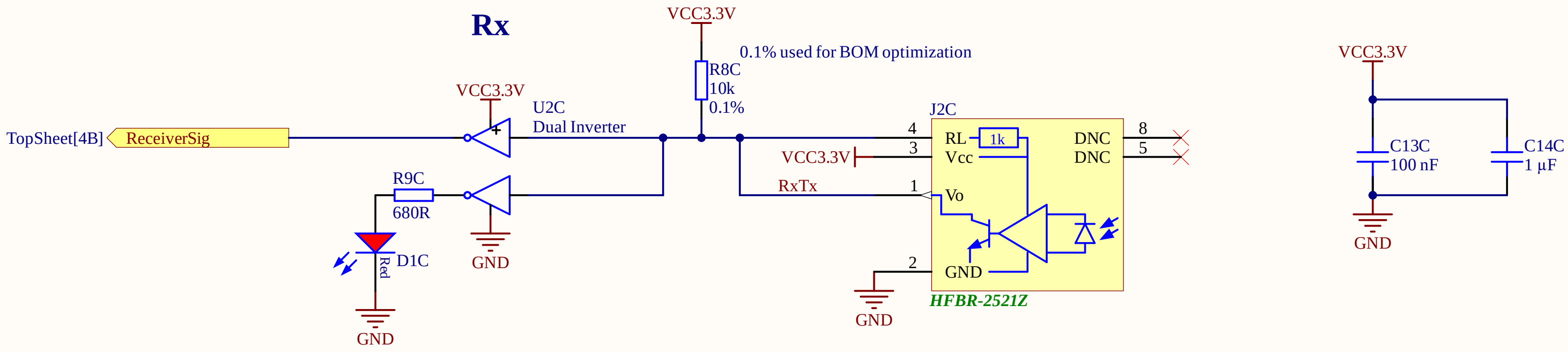
Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Receivers.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026 Sheet 5.2 of 20

Poor Mans Table of Truth for Rx

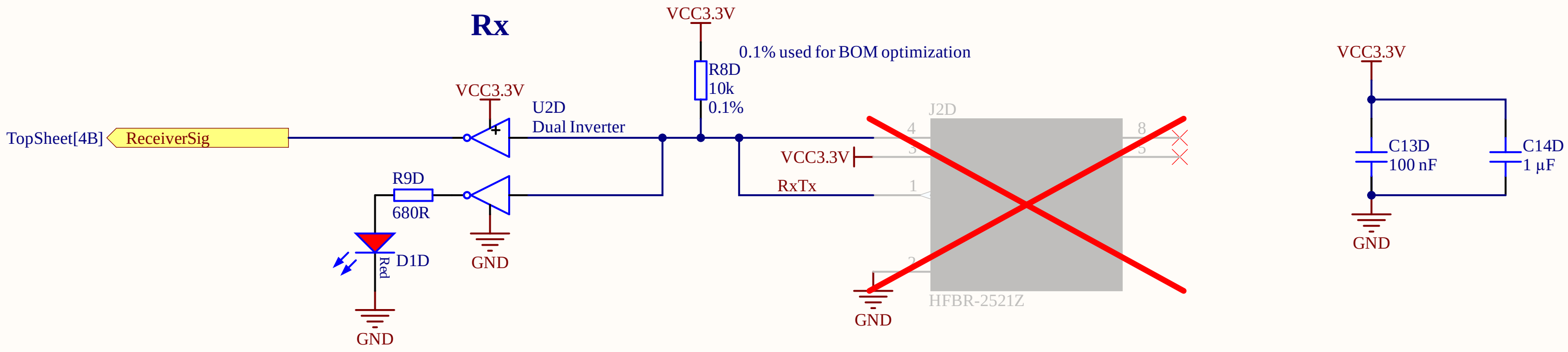
Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Receivers.SchDoc		UltraZohm
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026
		Sheet 5.3 of 20

Poor Mans Table of Truth for Rx

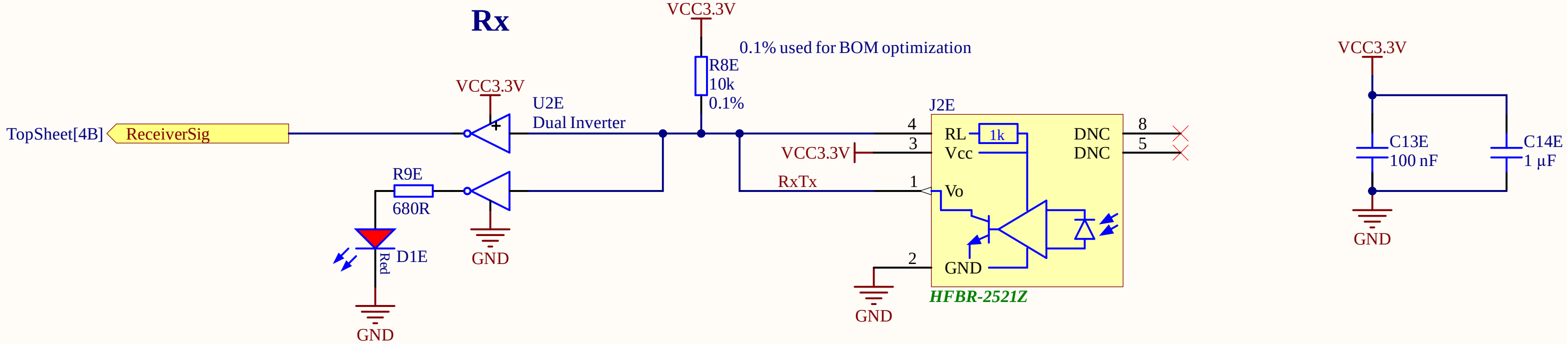
Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Receivers.SchDoc	**UltraZohm**
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Date: 23.06.2026 Sheet 5.4 of 20

Poor Mans Table of Truth for Rx

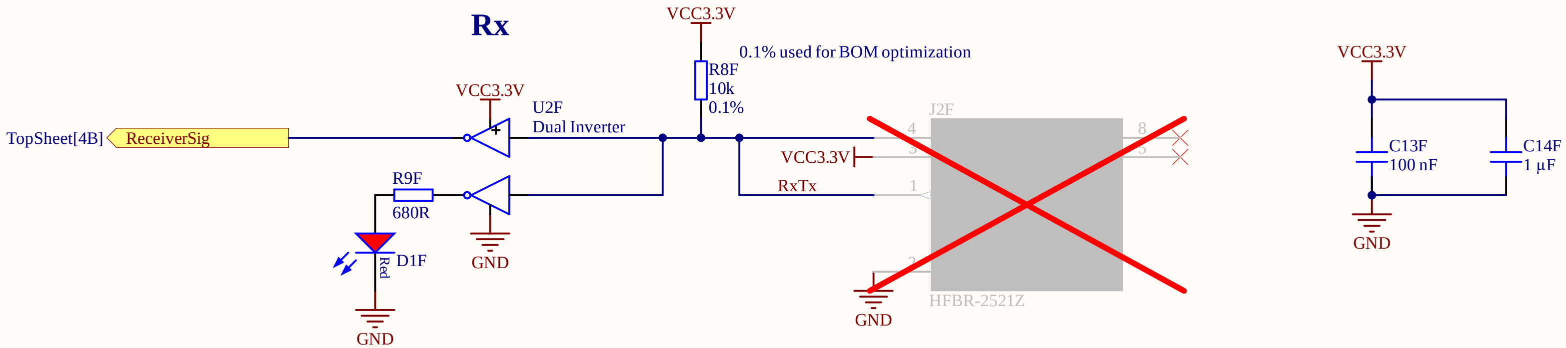
Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Receivers.SchDoc	UltraZohm
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Date: 23.06.2026 Sheet 5.5 of 20

Poor Mans Table of Truth for Rx

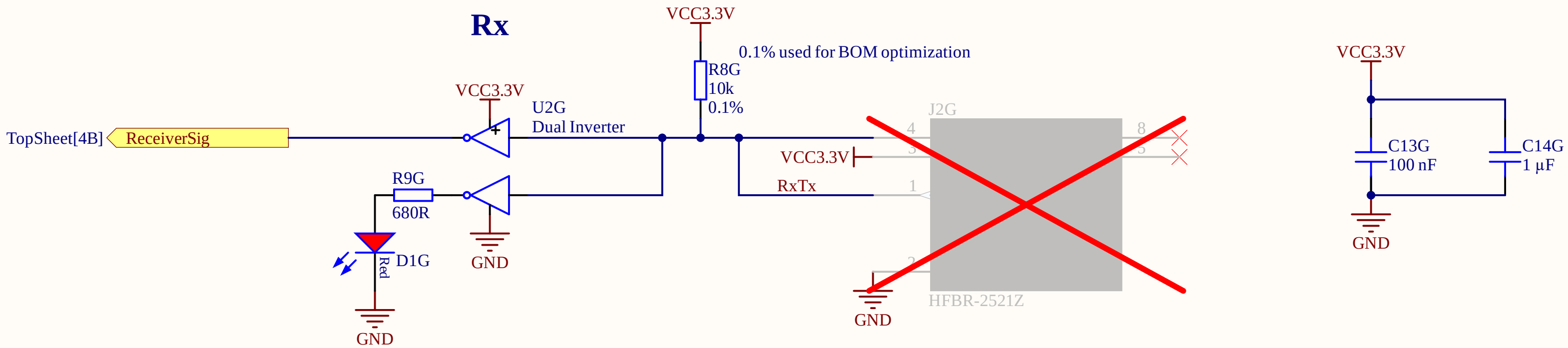
Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON



Title Receivers.SchDoc		UltraZohm
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026
		Sheet 5.6 of 20

Poor Mans Table of Truth for Rx

Receiver LED	Node RxTx	LED D2
OFF	HIGH	OFF
ON	LOW	ON

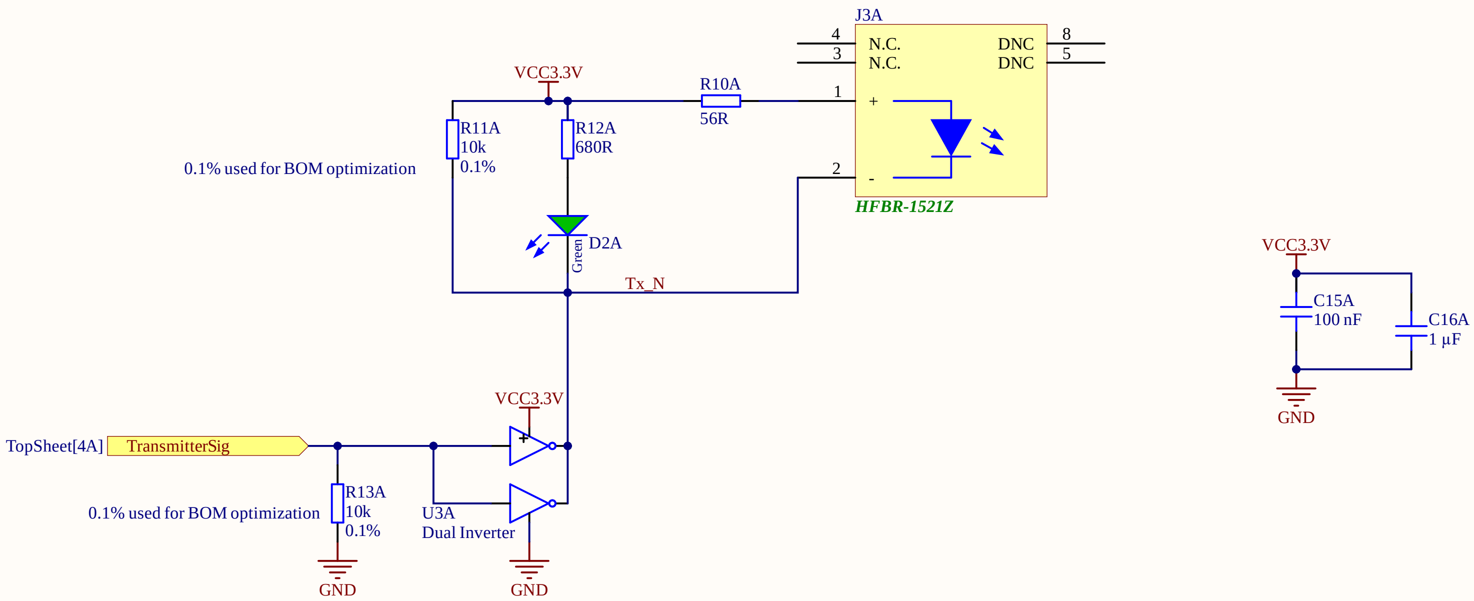


Title Receivers.SchDoc	UltraZohm
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Date: 23.06.2026 Sheet 5.7 of 20

Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Tx

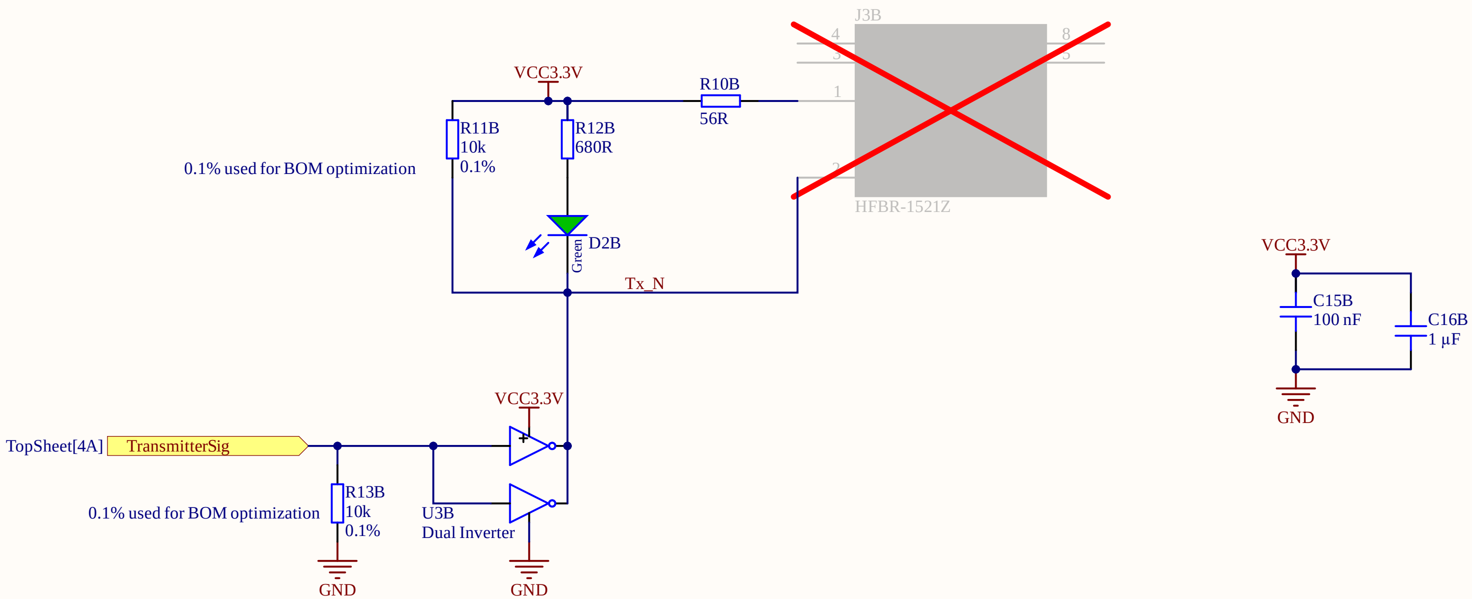


Title Transmitters.SchDoc	UltraZohm
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Date: 23.06.2026
	Sheet 6.1 of 20

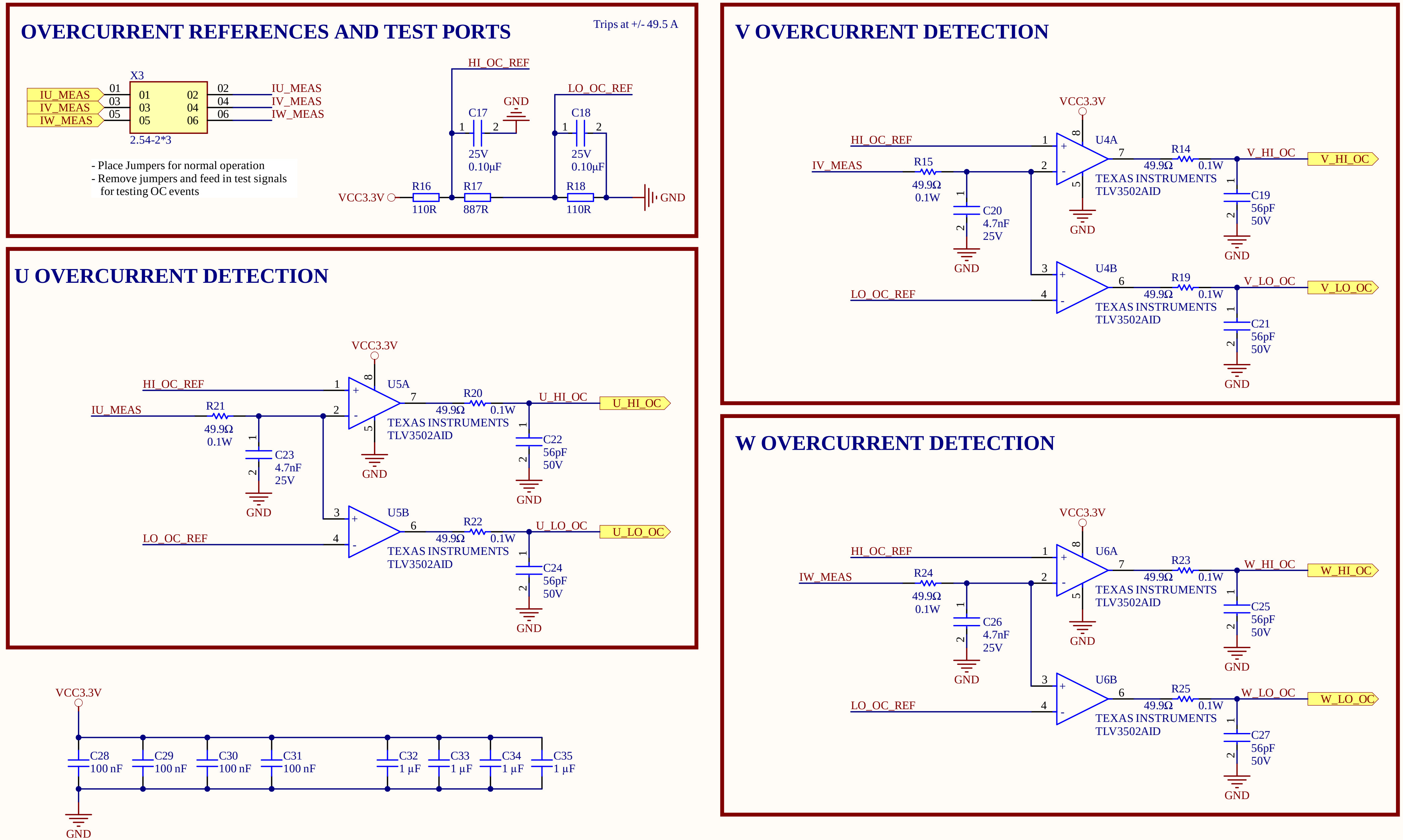
Poor Mans Table of Truth for Tx

Transmitter LED	Node Tx_N	LED D3	Signal
OFF	HIGH	OFF	LOW
ON	LOW	ON	HIGH

Tx



Title Transmitters.SchDoc	**UltraZohm** www.ultrazohm.com
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	Date: 23.06.2026
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Sheet 6.2 of 20



Title OvercurrentDetection.SchDoc

Revision: Rev04 Design Engineer: S. Park/M. Hoerner

Project: uz_per_wolfspeed_25kw_FM3.PrjPcb

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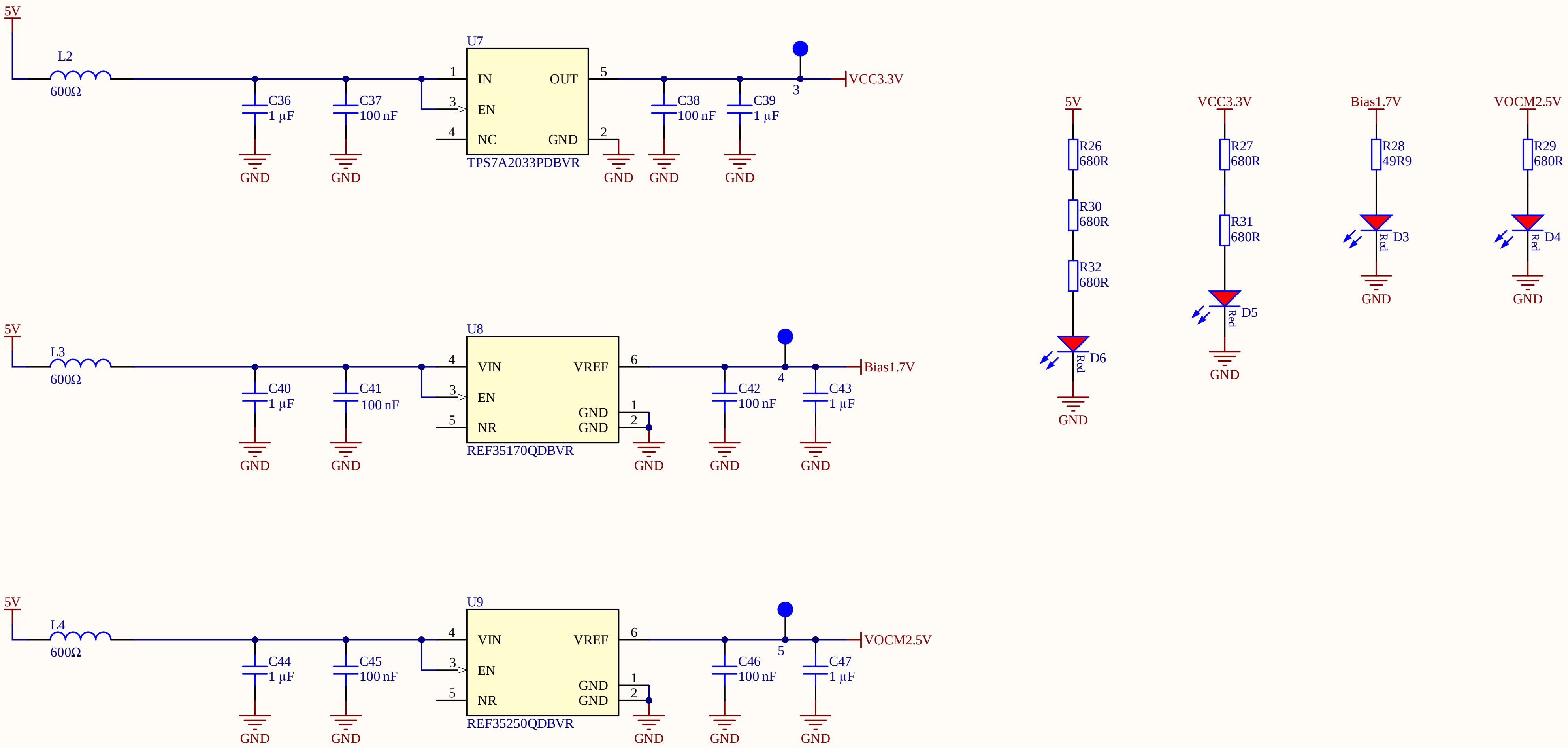
Date: 23.06.2026

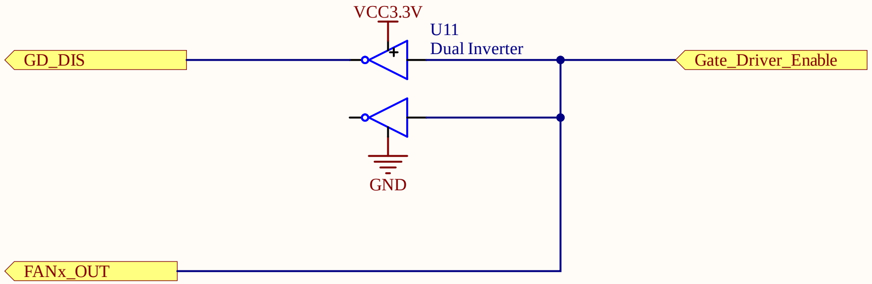
Sheet 7 of 20



Title Power.SchDoc		UltraZohm
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026
		Sheet 8 of 20

Invert Gate logic

- Gate_Driver_Enable high -> GD_DIS low -> Gates enabled
- Gate_Driver_Enable low -> GD_DIS high -> Gates disabled

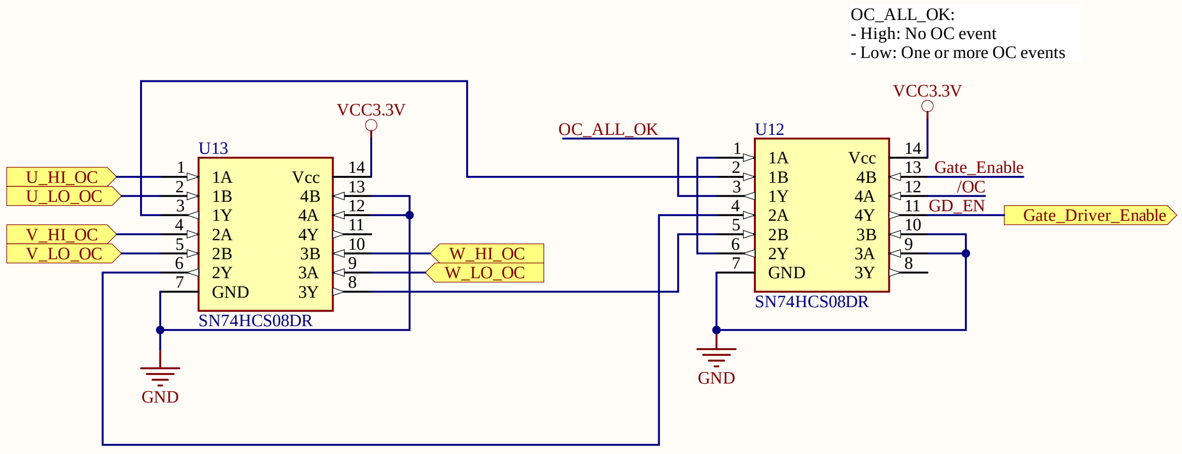


FANx_OUT logic

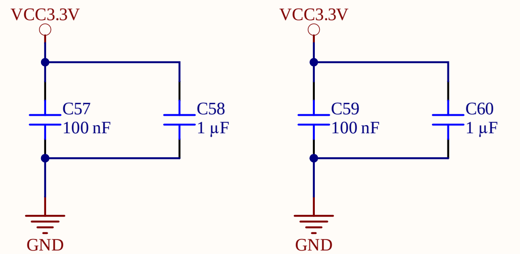
- Gate_Driver_Enable high -> FANx_OUT high -> Fan ON
- Gate_Driver_Enable low -> FANx_OUT low -> Fan OFF

Title GateAndFanControl.SchDoc		*UltraZohm*
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	www.ultrazohm.com
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026 Sheet 9 of 20

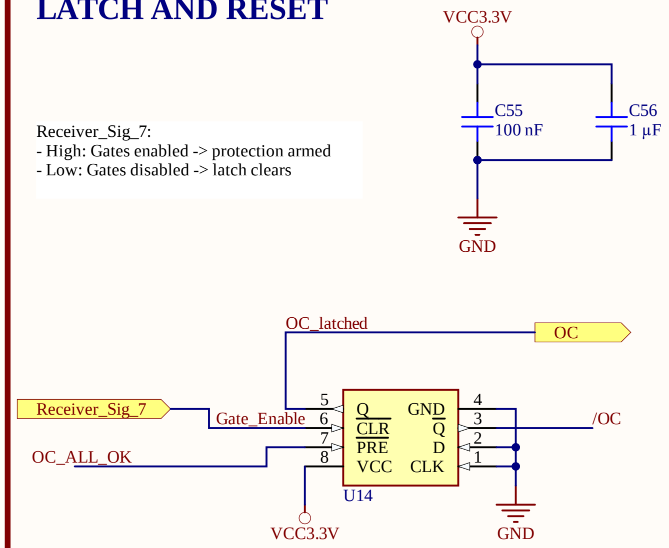
AND LOGIC



Gate_Enable	OC_latched	/Q = OC_latched_N	Gate_Driver_Enable
0	0	1	0
1	0	1	1
0	1	0	0
1	1	0	0

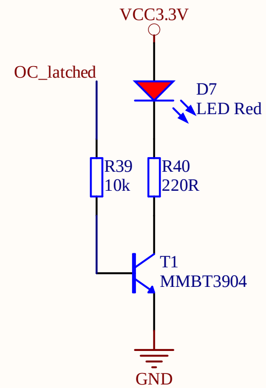


LATCH AND RESET



/PRE and /CLR must not be low simultaneously.

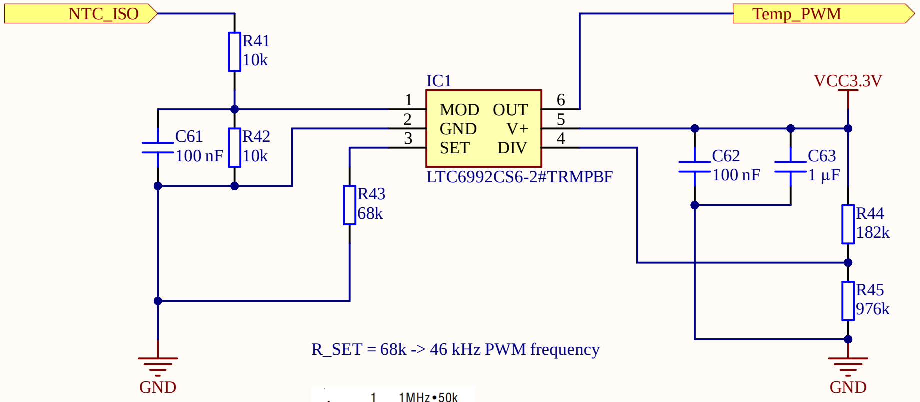
OC INDICATOR



Title OvercurrentDetectionLogic.SchDoc	UltraZohm www.ultrazohm.com
Revision: Rev04 Design Engineer: S. Park/M. Hoerner	Date: 23.06.2026
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb	Sheet 10 of 20

ANALOG NTC VALUE TO DIGITAL PWM CONVERSION

NTC_ISO assumed to output 0...2V. Voltage divider reduces to 0...1V



$$f_{OUT} = \frac{1}{t_{OUT}} = \frac{1\text{MHz} \cdot 50\text{k}}{N_{DIV} \cdot R_{SET}}$$

182/976 -> DIVCODE 2 -> N_DIV 16
POL = 1: Rising temperature at NTC leads to increasing duty cycle

Title TempSenseConditioning.SchDoc		UltraZohm www.ultrazohm.com
Revision: Rev04	Design Engineer: S. Park/M. Hoerner	
Project: uz_per_wolfspeed_25kw_FM3.PrjPcb		Date: 23.06.2026 Sheet 11 of 20